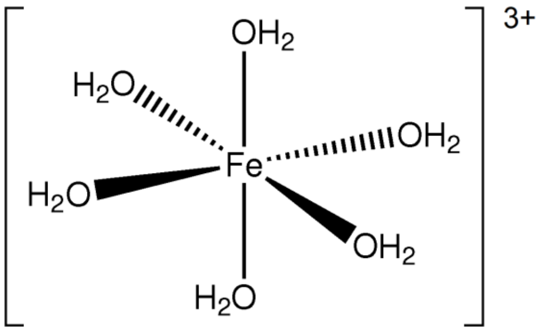
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 (4s^0)$ (1) partially-filled d-orbitals (1)	1	1		2		
	(b)			award (1) for either of following the energy of the (4s and) 3d-orbitals are all similar the ionisation energies to remove the (4s and) 3d-electrons are similar	1			1		
	(c)			 must show clear octahedral structure with bonds between Fe and O atoms in water	1			1		
	(d)	(i)		different ligands cause different amount of d-orbital splitting (1) so different frequencies/wavelengths of light are absorbed (and different frequencies/wavelengths are transmitted/reflected) (1)	1	1		2		
		(ii)		find a wavelength absorbed by $[Fe(H_2O)_6]^{3+}$ but not by $[Fe(H_2O)_5(OH)]^{2+}$ / any other species in the mixture		1		1		1

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(iii)		$K_c = \frac{[H^+][\{Fe(H_2O)_5(OH)\}^{2+}]}{[\{Fe(H_2O)_6\}^{3+}]}$ (1) unit ⇨ mol dm^{-3} (1)		1 1		2	1	
		(iv)		$[H^+]$ = 0.0282 mol dm^{-3} (1) $[Fe(H_2O)_6]^{3+} = \frac{0.103 \times 0.0282}{4.03 \times 10^{-3}}$ (1) $[Fe(H_2O)_6]^{3+}$ = 0.720 (1) mass of $FeCl_3.6H_2O$ = (0.720 + 0.103) × 270.4 = 222.5g (1)		1	3	4	3	
	(e)	(i)		$[Fe(H_2O)_3(OH)_3]$ or $Fe(OH)_3$ brown / red-brown / dark brown formula **and** colour needed	1			1		1
		(ii)		oxygen (from the air) can oxidise Fe^{2+} to Fe^{3+} (turning the precipitate brown) (1) award (1) for either of following • because oxygen has a more positive standard electrode potential than Fe^{3+} so it is a stronger oxidising agent • the EMF for the reaction between O_2 and Fe^{2+} is positive / +0.46V and positive reactions are feasible		1	1	2		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		electrode potential will be less positive / more negative (1) **must** attempt reason to gain this mark alkaline solution will reduce concentration of H^+ so equilibrium will move to left (1)			2	2		
	(f)	(i)		$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$		1		1		
		(ii)		CO has carbon in +2 oxidation state but the stable oxidation state of carbon is +4	1			1		
				Question 9 total	**6**	**8**	**6**	**20**	**4**	**3**